Smart Task Manager - README

1.Project Overview

The **Smart Task Manager** is a context-aware Java application designed for the SE3302 Software Construction course. It leverages real-time weather data to assist users in scheduling tasks efficiently, using Reactive Programming (Project Reactor) and a Swing-based GUI.

2. How to Run the App

To run the application, follow these steps from the project root directory (where the pom.xml file is located):

1. **Compile the project:**

   Bash

   ```
   mvn clean compile
   ```

2. **Execute the main application:**

   Bash

   ```
   mvn exec:java -Dexec.mainClass="taskmanager.MainApp"
   ```

3. Weather API Configuration

The application requires an API key from **OpenWeatherMap** to fetch live weather data.

- **Location:** Open the file src/main/java/taskmanager/MainApp.java.
- **Action:** Replace the placeholder string in the withWeatherApiKey("a885d3236fb957e3ab6abe72b776aabf" ") method with your valid API key.

4. Code Example: Using TaskManager API

Below is an example of how to programmatically interact with the core logic:

```java
TaskManager tm = TaskManager.builder()
        .withWeatherApiKey(apiKey: "a885d3236fb957e3ab6abe72b776aabf")
        .build();

// Add a couple of test tasks
// Seed initial data for testing system logic
Task task1 = new Task(
        "task-001",
        "Morning run",
        LocalDateTime.now().plusHours(hours: 2),
        true
);
Task task2 = new Task(
        "task-002",
        "Coding session",
        LocalDateTime.now().plusHours(hours: 4),
        false
);
// Populate manager state
tm.addTask(task1);
tm.addTask(task2);
// Verification log for system state
System.out.println("Tasks loaded: " + tm.getTasks().size());

// Wire this to the Swing UI
// UI Initialization and Dependency Injection
SmartTaskManagerFrame frame = new SmartTaskManagerFrame(tm);
// Ensure UI thread safety via invokeLater
javax.swing.SwingUtilities.invokeLater(() -> frame.setVisible(true));
```

5. Team Members

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